THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 7 – Black-Scholes-Merton Model

Dr Xiaoting Wei

Lecture Outline

Assumptions about how stock prices evolve

Volatility

Assumptions underlying BSM model

Theory behind BSM

The BSM model

Link between BSM and Binomial model

Implied volatility

Volatility smile

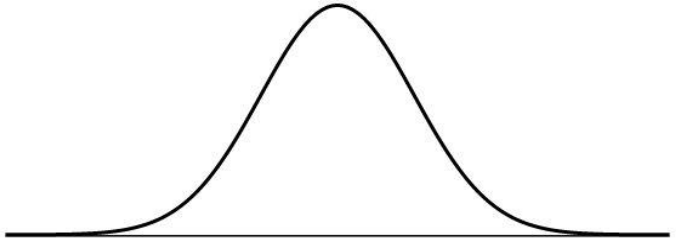
The VIX Index

BSM for Futures Options

Readings: Chapters 13, 19.1 & 19.2

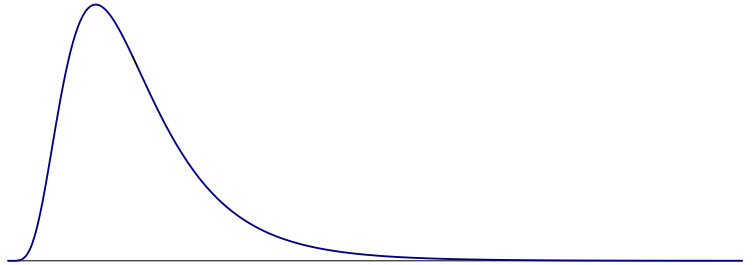
Assumptions about how stock prices evolve

Normal distribution



$X \sim \varphi(m, v)$

Lognormal distribution



For a lognormal distribution, X does not follow a normal distribution. Instead, ln(X) follows a normal distribution.

$\ln(X) \sim \varphi(m, v)$

Assumptions about how stock prices evolve

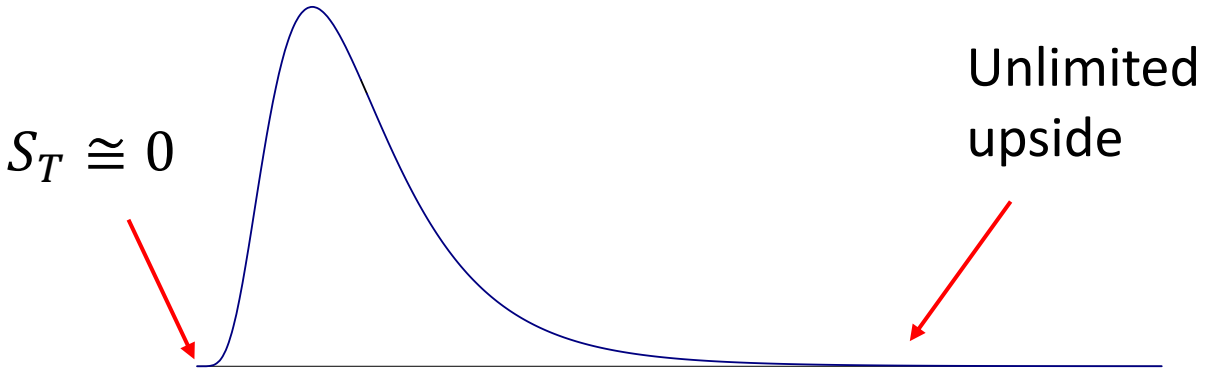
The Black-Scholes-Merton (BSM) model assumes that the return on the stock in a short period of time is normally distributed.

- μ: Expected return on the stock
- σ: Volatility of the stock return
- $\frac{\Delta S}{S} \sim \varphi(\mu \Delta t, \sigma^2 \Delta t)$, where $\frac{\Delta S}{S}$ is the stock return over a small time period of Δt.
- Notice that $\mu \Delta t$ is an arithmetic mean (i.e., not a continuously compounded return). It would not matter for a very short period of time, but for a long period of T, we need a continuously compounded return $\ln(S_T/S_0)$ for the BSM model.
- $\ln \frac{S_T}{S_0} \sim \varphi \left[\left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right]$
- $\ln S_T \sim \varphi \left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right]$

Assumptions about how stock prices evolve

$$\ln S_T \sim \varphi\left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]$$

- $\ln S_T$ follows a normal distribution, and therefore S_T follows a lognormal distribution.
- It makes intuitive sense that a stock price can rise significantly but cannot fall below zero.



Assumptions about how stock prices evolve

The expected value (or mean) of S_T:

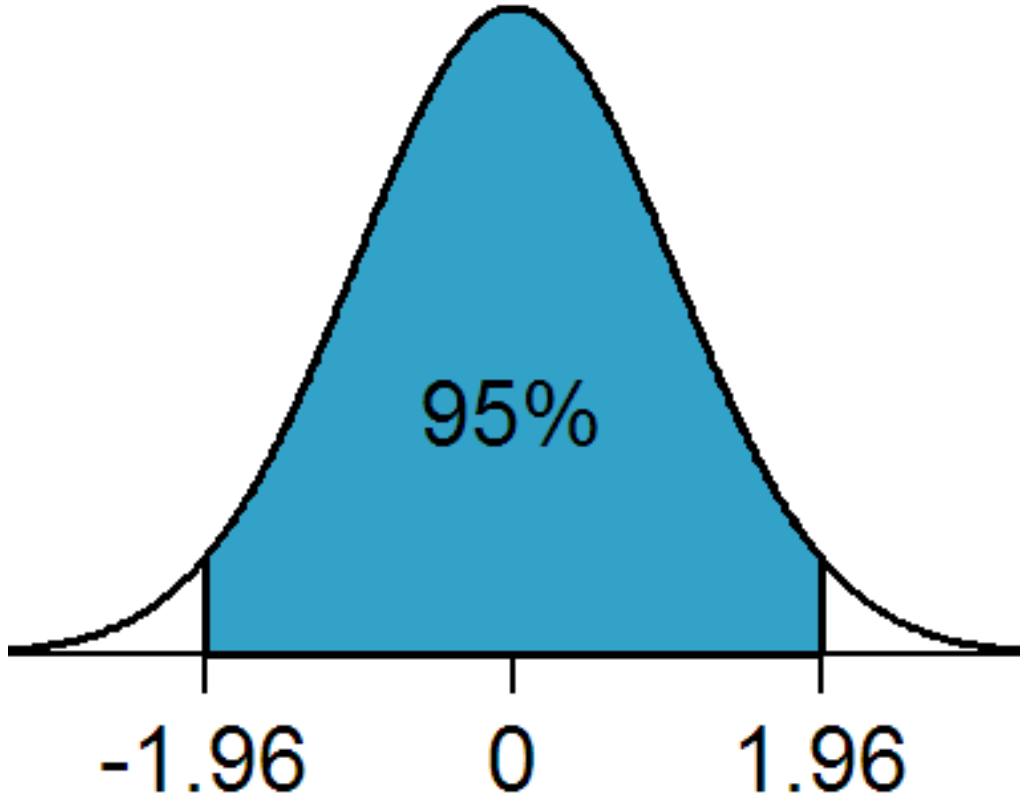
$$E(S_T) = S_0 e^{\mu T}$$

The variance of S_T:

$$\text{var}(S_T) = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right)$$

Assumptions about how stock prices evolve

Example: Find mean, variance, and the 95% confidence interval for the stock price in six months. The stock has a current price of $50, an expected return of 20% per annum, and a volatility of 30% per annum.



Assumptions about how stock prices evolve

The probability distribution of the stock price in six months is given by:

$$lnS_T \sim \varphi\left[ln50 + \left(0.2 - \frac{0.3^2}{2}\right)0.5, (0.3\sqrt{0.5})^2\right]$$

$$lnS_T \sim \varphi[3.990, 0.045]$$

There is a 95% probability that a normally distributed variable has a value within 1.96 standard deviations of its mean. Hence, with 95% confidence:

$$3.990 - 1.96 \times \sqrt{0.045} < \ln S_T < 3.990 + 1.96 \times \sqrt{0.045}$$

This implies:

$$e^{3.990-1.96\times0.212} < S_T < e^{3.990+1.96\times0.212}$$

$$35.676 < S_T < 81.902$$

Assumptions about how stock prices evolve

Thus, there is a 95% probability that the stock price in six months time will lie between 35.676 and 81.902.

The mean of S_T is:

$$50e^{0.2\times0.5} = 55.259$$

The variance of S_T is:

$$50^2 e^{2\times0.2\times0.5}\left(e^{0.3^2\times0.5} - 1\right) = 140.546$$

Volatility

Volatility is a measure of our uncertainty about stock returns.

The volatility of the stock is defined as the standard deviation of the return provided by the stock in one year.

- This return is expressed using continuous compounding.

As shown in the previous slides, the volatility of a stock over a particular period of time can be expressed as:

$$\sigma\sqrt{T}$$

Volatility

Example: Suppose that a stock's current price is \$70, and that σ = 0.2 or 20% per annum. Calculate a one-standard-deviation move in the stock price in one fortnight.

- The standard deviation of the percentage change in the stock price in one fortnight is approximately:

$$20 \times \sqrt{\frac{1}{26}} = 3.922\%$$

- A one-standard-deviation move in the stock price in one fortnight is therefore:

$$70 \times 0.03922 = \$2.75$$

Assumptions underlying BSM

1. Stock price behaviour corresponds to the lognormal model we discussed earlier in the lecture with μ and σ constant.
2. There are no transaction costs or taxes.
3. There are no dividends on the stock during the life of the option.
4. There are no riskless arbitrage opportunities.
5. Security trading takes place continuously in time.
6. Investors can borrow and lend at the same risk-free rate.
7. The short-term risk-free rate of interest r is constant.

Theory behind BSM

We will not cover a complete derivation of BSM. Rather, below we discuss the intuition behind the model.

The arguments used here are analogous to the no-arbitrage arguments used under the binomial model.

Take a riskless portfolio comprising a position in the option and a position in the underlying stock.

If there are no arbitrage opportunities, the return for the portfolio must be the risk-free interest rate.

The reason we can set up this riskless portfolio is that the stock price and option price are both affected by the same underlying uncertainty, i.e. stock price movements.

Theory behind BSM

Thus in any **short period** of time:

- The price of a call option has a linear positive relationship with the price of the underlying stock; and,
- The price of a put option has a linear negative relationship with the price of the underlying stock.
  - In both of these cases, when an appropriate portfolio of the stock and the option is set up, the gain or loss from the stock position will be completely offset by a corresponding loss or gain on the option position so that the overall value of the portfolio at the end of the period is known with certainty.
  - As the overall value is known with certainty, and the gains/losses are completely offset, the return from the riskless portfolio in any short period of time must be the risk-free interest rate.
  - This is the key element in the BSM model that leads to the pricing formulae.

Theory behind BSM

Suppose, for example, that at some point in time the relationship between a small change in the stock price, ΔS , and the resultant small change in the price of a European call option, Δc , is given by:

$$\Delta c = 0.4 \Delta S$$

This means a riskless portfolio would consist of:

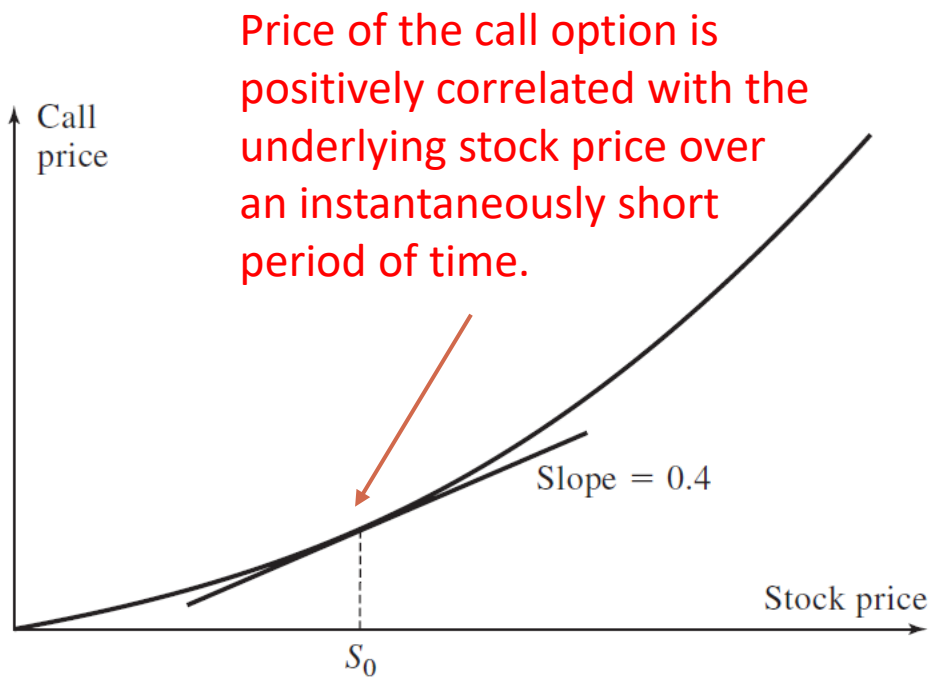
- a long position in 40 shares
- a short position in 100 call options.

Suppose that the stock price increases by 10 cents. The option price will increase by 4 cents and the $40 \times 0.10 = \$4$ gain on the shares is equal to the $100 \times 0.04 = \$4$ loss on the short option position.

Theory behind BSM

There is one important difference between the analysis here and the analysis using a binomial model. Here, the riskless position only lasts for an instantaneously short period of time. To remain riskless, it must be frequently adjusted or rebalanced.

For example, the relationship between Δc and ΔS might change from $\Delta c = 0.4\,\Delta S$ today to $\Delta c = 0.5\,\Delta S$ tomorrow.



The BSM model

The BSM formula to value a European option written on a non-dividend paying stock is:

$$c = S_0\, N(d_1) - \mathrm{K}\, e^{-rT} N(d_2)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

Where: $$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

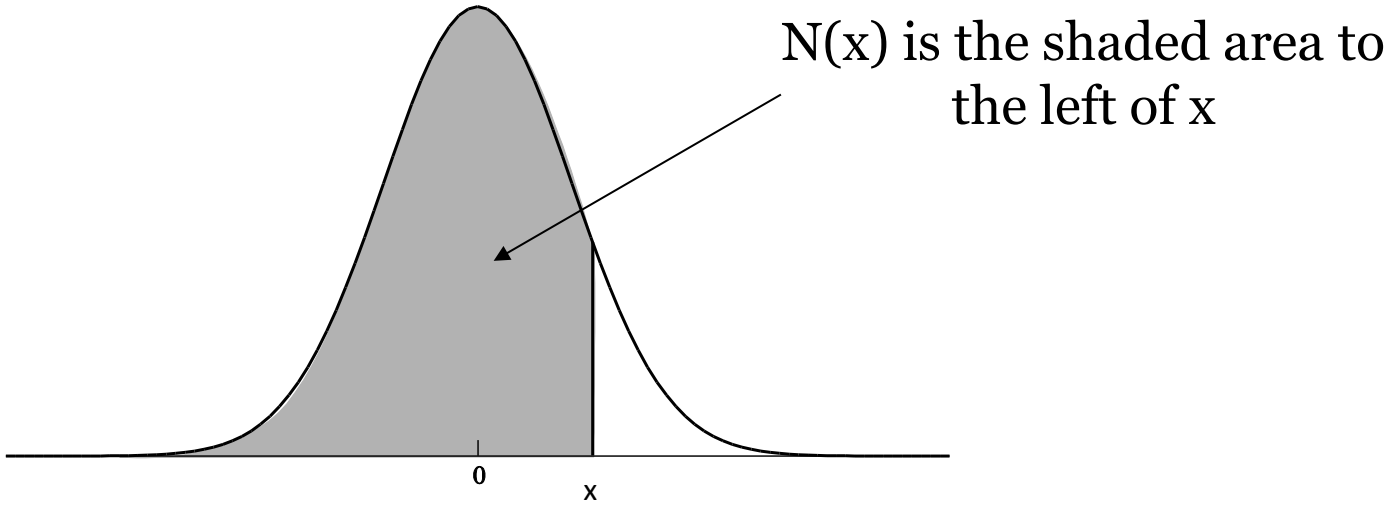
The BSM model

What are d_1, d_2 and $N(x)$?

- d_1 and d_2 are just preliminary calculations.
- $N(x)$ represents the cumulative probability function for a standardised normal variable. It is the probability that a variable with a standard normal distribution will be less than x. This value can be obtained from the probability tables on pp. 600-601 of the textbook (see LMS for a copy of this).
- $N(d_2)$ is the probability that the call option will be exercised.
- $N(d_1)$ is related to delta hedging – to be covered in Week 10.

The BSM model

X in the standard normal distribution table is a standard normal random variable which has a mean of 0 and a standard deviation of 1.



The BSM model

$$c = S_0\, N(d_1) - \mathrm{K}\, e^{-rT} N(d_2)$$

Where: $d_1 = \dfrac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

As $S_0 \to 0$,
d_1 & d_2 large -ve
$N(d_1), N(d_2) \to 0$
$C \to 0$

As $S_0 \uparrow$, d_1 & $d_2 \uparrow$, $N(d_1)$ & $N(d_2) \to 1$
$C \to S - Ke^{-rT}$

Value of forward with no income (lecture 3),
Call almost certainly exercised when deep in $

The BSM model

$$p = K\, e^{-rT}\, N(-d_2) - S_0\, N(-d_1)$$

Where: $d_1 = \dfrac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

As $S \uparrow$,

d_1 & $d_2 \uparrow$

$N(-d_1), N(-d_2) \to 0$

$P \to 0$

As $S \to 0$,

d_1 & d_2 large -ve

$N(-d_1), N(-d_2) \to 1$

$P \to Ke^{-rT} - S$

Formula for a short forward

- Derivation uses put call parity + BSM call formula (tutorial)

The BSM model

Woodside Petroleum (WPL) share price (S_0) is \$30.

The standard deviation (σ) of WPL returns is 30% pa.

Riskless interest rate (r) is 4% pa.

The call option has a \$25 strike (K) and time to expiry (T) of six months.

Calculate the Black-Scholes price of this call option.

The BSM model

$$d_1 = \frac{ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$= \frac{ln(30/25) + (0.04 + 0.3^2/2)0.5}{0.3\sqrt{0.5}}$$

$$= 1.0598 \approx 1.06$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$= 1.0598 - 0.3\sqrt{0.5}$$

$$= 0.85$$

$$N(d_1) = 0.8554$$

$$N(d_2) = 0.8023$$

$$c = S_0 N(d_1) - K e^{-r \cdot T} N(d_2)$$

$$= 30N(1.06) - 25e^{-0.04 \times 0.5} N(0.85)$$

$$= 30 \times 0.8554 - 25(0.9802)(0.8023)$$

$$= \$6.00$$

The BSM model

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

The BSM model

A 7-month European put option on the stock of Telstra has an exercise price of $4.80.

The current stock price is $4.50,

The risk-free rate is 8% per annum, and

The volatility is 30% per annum.

Calculate the Black-Scholes price of this put option.

The BSM model

$$
\begin{aligned}
d_1 &= \frac{ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \\
&= \frac{ln(4.5/4.8) + (0.08 + 0.3^2/2)\frac{7}{12}}{0.3\sqrt{\frac{7}{12}}} \\
&= 0.0366 \approx 0.04
\end{aligned}
$$

$$
\begin{aligned}
d_2 &= d_1 - \sigma\sqrt{T} \\
&= 0.0366 - 0.3\sqrt{\frac{7}{12}} \\
&= -0.19
\end{aligned}
$$

The BSM model

$$-d_1 = -0.04$$
$$-d_2 = -(-0.19) = 0.19$$
$$N(-d_1) = 0.4840$$
$$N(-d_2) = 0.5753$$

$$\begin{aligned} p &= Ke^{-rT}N(-d_2) - S_0N(-d_1) \\ &= 4.80e^{-0.08*\frac{7}{12}} \times (0.5753) - 4.50 \times (0.4840) \\ &= 0.4575 \end{aligned}$$

The BSM model

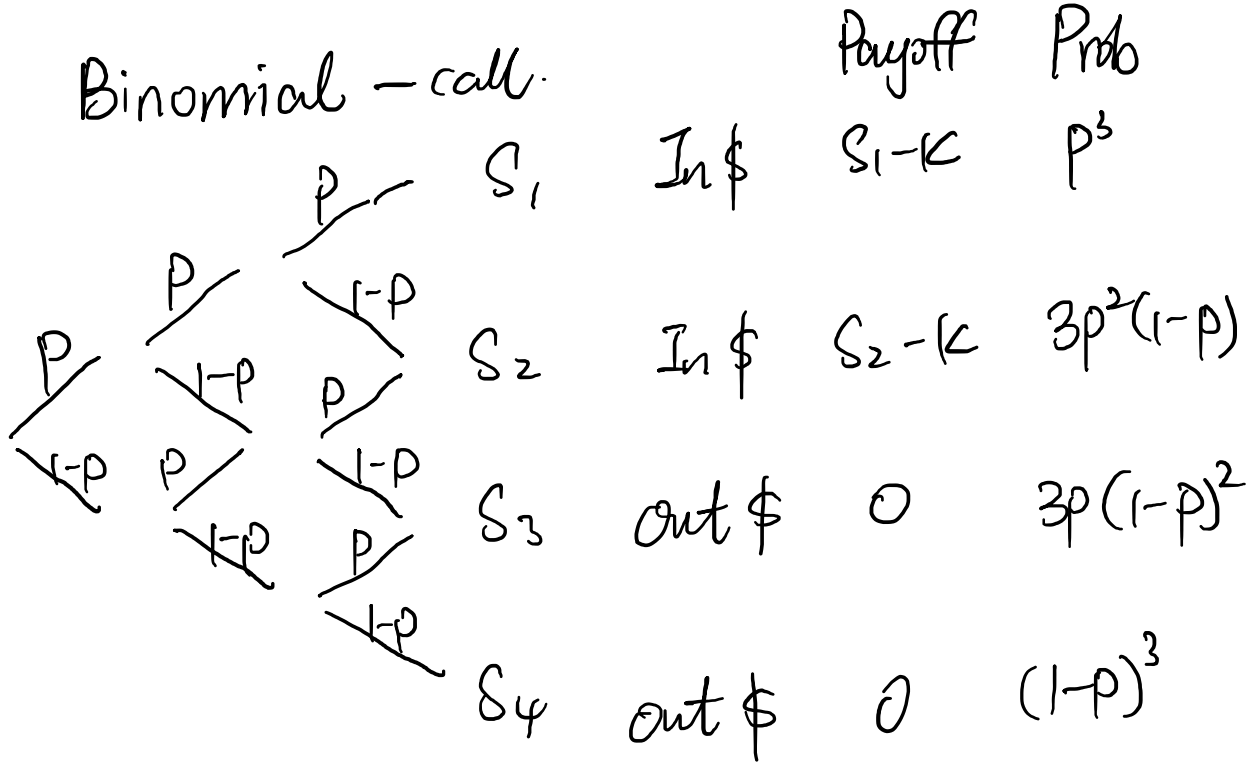
x	.00	.01	.02	.03	.04	.05	.06	.07	.08	. 09
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367

The BSM model

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
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0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
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0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
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1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

Link between BSM and Binomial model

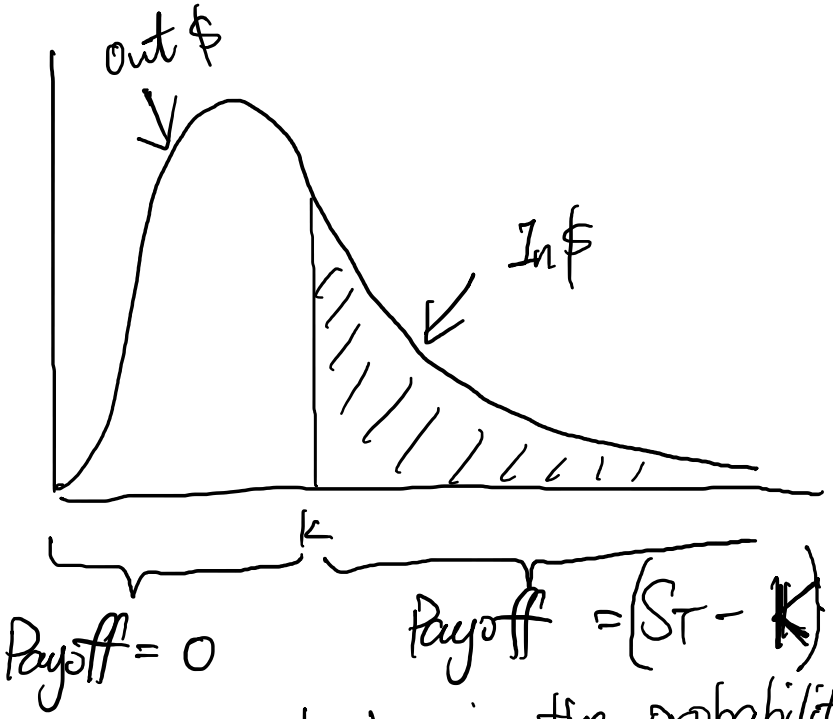
Binomial – call.



Exp. Payoff = $p^3(S_1-K) + 3p^2(1-p)\cdot(S_2-K)$

C = Exp. Payoff $\cdot\ e^{-rT}$

BSM



$N(d_2)$ is the probability that $S_T > K$

Link between BSM and Binomial model

Getting behind the formula

Prob of in $

Payoff at T out $

Expected payoff at T in $

Prob out of $

$$c = e^{-rT}\left\{N(d_2)\left[S_0 e^{rT}\left(\frac{N(d_1)}{N(d_2)}\right) - K\right] + 0 \times \left[1 - N(d_2)\right]\right\}$$

Expectation at T over entire distribution

$\frac{N(d_1)}{N(d_2)} > 1$ scaling, so the expectation relates to stock prices at T when call in the $

Expand and simplify

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

Link between BSM and Binomial model

Now let's use the simulator…

Check the question on slide 22. Using BSM, c=6.00

What is the option price using a 2-step Binomial model? c=6.19

What about 5-step? c=6.07

What about 20-step? c=6.01

Implied volatility

- Cannot directly observe σ, must be estimated
- Recall $C\ (S_0,\ K,\ T,\ r,\ \sigma)$
- Given $C,\ S_0,\ K,\ T,\ r,$ implied vol is the value of σ which substituted in the BSM formula, gives
  - theoretical option price = current market price.
- Reflects markets assessment of future stock volatility over the life of the option.
- Solved iteratively as BSM formula cannot be inverted.

Implied volatility

Example: Assume value of 3-month European call on a non dividend paying stock is $2.10. If the current stock price is $21, strike is $20 and risk-free rate is 10% p.a., what is the implied volatility of the option

Solve iteratively

σ=0.20, C=$1	(vol too low)
σ=0.40, C=$4	(vol too high)
σ=0.30, C=$2.50	(vol too high)

Keep iterating until C=$2.10

Implied volatility

Where Does the Option Market Price Come From?

Buyers and sellers trade options based on their expectations and market conditions.

Their decisions reflect factors such as:

- Expected future volatility
- Supply and demand
- Hedging needs
- Market uncertainty and upcoming events
- Liquidity and risk preferences

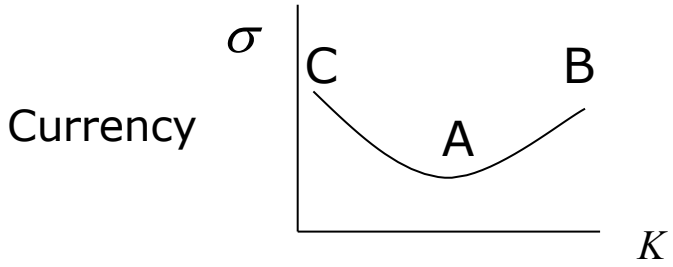
The interaction of **buying and selling** determines the option's market price.

We then **work out the implied volatility** from the market price.

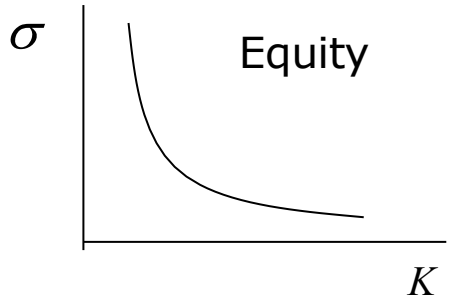
Market trading → Option price → Implied volatility

Volatility smiles

- If BSM model is perfect, all options on same stock would generate same implied vol
- In practice, holding everything else constant, there are patterns in implied vol as a function of
  - Strike price: vol smiles (currency), vol smirk (equity)



A = at-the-money options
B = out of $ calls, higher prob of large increase in S than N dist'n, higher call value, higher implied vol
C = out of $ puts, higher prob of large decrease in S than N dist'n, higher put value, higher implied vol

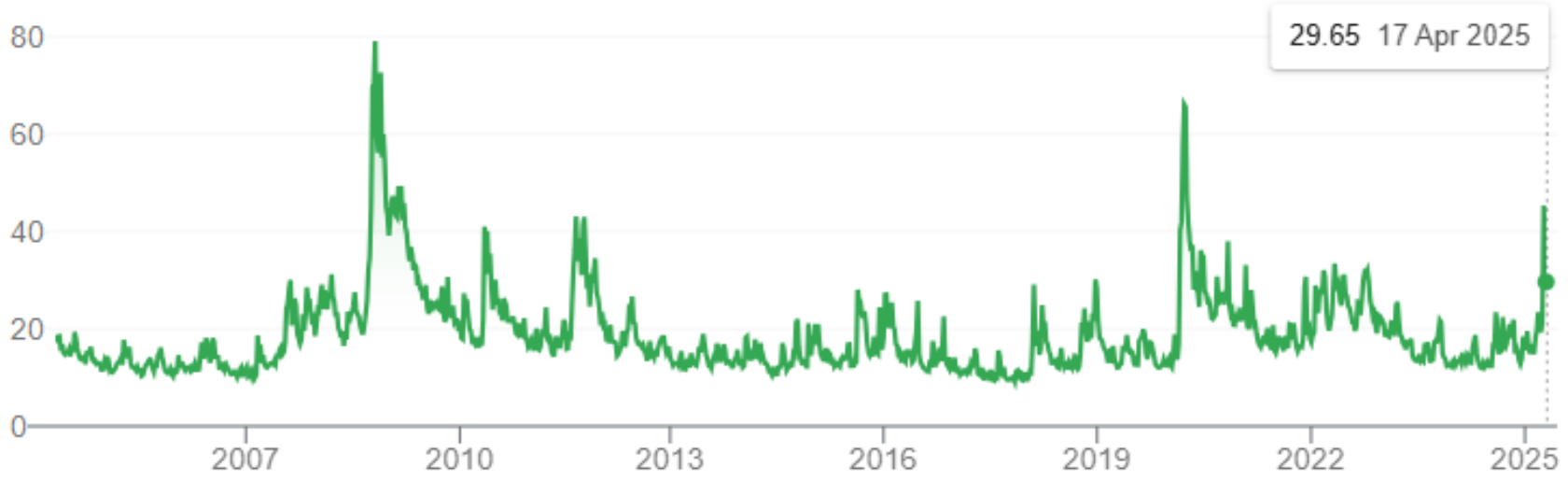


Negative skewness in stock returns, higher prob of large price decrease

The VIX Index

VIX is an index of the implied volatility of 30-day options on the S&P 500 calculated from a wide range of calls and puts.

An index value of 15 indicates that the calculated implied volatility of 30-day options on the S&P 500 is 15% pa.



The VIX Index

There are futures on the VIX and options on the VIX. Below is an illustration of futures on the VIX.

- A trader buys an April futures contract on the VIX when the futures price is 18.5 (corresponding to a 30-day S&P 500 volatility of 18.5% pa) and closes out the contract when the futures price is 19.3 (corresponding to a 30-day S&P 500 volatility of 19.3% pa). One contract is on 1000 times the index.
- The trader makes a gain of $(19.3 - 18.5) \times 1,000 = \$800$

BSM for Futures Options

Replace S_0 with $F_0 e^{-rT}$

$$c = e^{-rT}[F_0 N(d_1) - KN(d_2)]$$
$$p = e^{-rT}[KN(-d_2) - F_0 N(-d_1)]$$

Where:

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}$$
$$d_2 = \frac{\ln(F_0/K) - \sigma^2 T/2}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

BSM for Futures Options

Example

- Consider a European put futures option on gold.
- The time to maturity is six months, the current futures price is $400 per ounce, the exercise price is $380 per ounce, the risk-free interest rate is 10% p.a., and the volatility of the futures price is 30% p.a.
- What should be the price of the European put futures option?

BSM for Futures Options

$$d_1 = \frac{\ln(400/380) + 0.3^2 \times 6/12/2}{0.3\sqrt{6/12}} = 0.3479 \approx 0.35$$

$$d_2 = 0.3479 - 0.3\sqrt{6/12} = 0.14$$

So:

$N(-d_1) = 0.3632$

$N(-d_2) = 0.4443$

The put price is:

$p = e^{-0.1\times0.5}(380 \times 0.4443 - 400 \times 0.3632) = 22.41$

Next Week

We are going to look at how **dividends** affect all the option pricing models we have learnt so far…